

C# Syntax at a Glance

Introduction

This guide provides a step-by-step approach to understanding and using fundamental C# syntax. It covers essential components, including data types, operators, program structure, and control structures, to help you build functional .NET applications.

Declare Data Types and Variables

- Choose a data type: Start by selecting a data type, such as `int` for integers, `string` for text, `double` for decimals, or `bool` for true/false values.
- Declare a variable: Follow the syntax `dataType variableName = value;`. Example: `int age = 25;` declares an integer variable `age` with a value of 25.

Use Operators and Create Expressions

- Arithmetic operators: Perform calculations using `+` (add), `-` (subtract), `*` (multiply), and `/` (divide). Example: `int total = 5 + 3;` stores the result (8) in `total`.
- Logical operators: Use `&&` (AND) and `||` (OR) in conditions to combine multiple requirements. Example: `if (isMember && hasCoupon)` checks if both conditions are true.
- Assignment operator: Use `=` to assign values to variables. Example: `int score = 10;`

Organize Code with Language Structure

- Create a namespace: Start with a namespace to encapsulate code, followed by `{}` brackets to contain your program.

- Example:

```
namespace MyFirstApp { /* code here */ }
```

- Define a class: Inside the namespace, declare a class to organize your program's code.

- Example:

```
class Program { /* methods go here */ }
```

- Set the entry point with main: Add static void `Main(string[] args) { /* code */ }` as the starting point for program execution.

Write Control Structures

- If-Else statements: Decide actions based on conditions using if and else.

```
if (userInput == "password") { access = true; } else { access = false; }
```

- Switch statements: Simplify complex conditionals with switch.

```
switch (option) { case "home": page = "Home"; break; default: page = "Not  
found"; break; }
```

Conclusion

This guide gives you the basics to start coding in C#. Follow these steps as a foundation to create well-structured, functional C# applications. With practice, these steps will become a natural part of your coding process.